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Generative AI in graduate bioprocess engineering exams: Is attention all students need?

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ABSTRACT

State-of-the-art large language models (LLMs) can now answer conceptual textbook questions with near-perfect accuracy and perform complex equation derivations, raising significant concerns for higher education. This study evaluates the performance of LLMs on graduate-level bioprocess engineering exams, which include multiple-choice, short-answer, and long-form questions requiring calculations. First, allowing students to use LLMs led to a 36 % average score increase compared to exams taken with only textbooks and notes. Second, as students gained more experience using LLMs, their performance improved further, particularly among students with disabilities. Third, under optimized conditions on two exams, OpenAI's GPT-4o scored approximately 70 out of 100, while more advanced models, such as OpenAI o1, o3, GPT-4.5, Qwen3–235B-A22B, and DeepSeek R1, scored above 84, outperforming 96 % of human test-takers. This indicates that students with access to more capable AI tools may gain an unfair advantage. Fourth, we propose guidelines for developing exam questions that are less susceptible to LLM-generated solutions. These include tasks such as interpreting graphical biological pathways, answering negatively worded conceptual questions, performing complex numerical calculations and optimizations, and solving open-ended research problems that demand critical thinking. This article calls for urgent reforms to bioprocess engineering education, advocating for the integration of LLM literacy through hands-on activities that address both practical applications and ethical considerations.

1. Introduction

Large Language Models (LLMs) are a subdomain of Artificial Intelligence (AI). LLMs like OpenAI's GPT-4o (OpenAI, 2023), o1, Qwen (Yang et al., 2025; Yang et al., 2024), and DeepSeek R1 (Guo et al., 2025), have significantly infiltrated higher education. Recent studies have highlighted the multifaceted impact of AI across various academic settings. For example, the emergent role of large language models (LLMs) in higher education and research has been discussed (Alqahtani et al., 2023). Additionally, recent research explores the opportunities and challenges that LLMs present for education, addressing both benefits (e.g., enhancing student support) as well as ethical concerns (Kasneci et al., 2023). Similarly, an examination of the implications of ChatGPT and other LLMs in academia reveals their potential to enhance or disrupt traditional educational practices (Meyer et al., 2023). In the era of generative AI, instructors must reconsider the weighting factors for homework and take-home exams/projects. Further studies investigated the practical applications, ethical considerations, and future directions of LLMs in education (Abd-Alrazaq et al., 2023; Baskara, 2023; Dai et al. 2023; Jeon and Lee, 2023; Perkins, 2023; Yan et al., 2024). In the field of chemical engineering, integrating LLM into chemical engineering core courses has been explored (Tsai et al., 2023). Recent study and survey have tested GPT-3.5 as a useful tool for chemical engineering lab session, highlighting its pitfalls, challenges and opportunities (Keith et al., 2025). In medical education, LLMs can generate customized study plans and explanations based on a student's strengths and weaknesses, promoting more efficient and effective learning (Abd-Alrazaq et al., 2023; Scherr et al., 2023; Sriram et al., 2025). Other studies covered practical teaching issues like assessment integrity and plagiarism detection (Bernabei et al., 2023; Nikolic et al., 2023).

Despite a wide range of studies on ethical and responsible incorporation of LLMs into higher education curricula, there remains gaps in the literature. First, most existing studies focus on qualitative assessments, or theoretical frameworks without providing data on how state-of-art

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LLMs (such as OpenAI o1, GPT-4.5, or Deepseek R1) perform in real teaching settings. Second, graduate level course work involves more critical thinking, data interpretation, and reasoning. Although some educators suggested that ChatGPT can offer teaching assistance in engineering education (Filippi and Motyl, 2024), few studies have investigated how LLMs resolve exam questions from bioprocess engineering courses, neither do we know the outcomes if students have been offered with newest LLMs for their course work and exams. To address this, our study offers a detailed quantitative analysis of LLMs in a graduate-level bioprocess engineering exam. We evaluate its performance in solving textbook and exam questions, assess the reliability of AI grading, and propose strategies to design questions that challenge LLMs. Although one effective method to minimize the impact of LLMs in exams is reverting to traditional paper-and-pencil, this method seems to be on the wrong side of history. It is necessary to embrace these technologies and prepare students for an era dominated by LLMs. In this study, we aim to provide actionable insights and an example course module about how to integrate LLMs into higher education. The course (Bioprocess engineering) included 24 graduate students with diverse backgrounds from environmental/chemical engineering to bioengineering. We also enrolled 4 high school students to participate in this study by using LLMs to solve course exams. One key motivation for this study is the need to redesign homework and take-home exam questions. As LLMs become more sophisticated, traditional question formats may no longer be effective in assessing students. Another important consideration is the need to better accommodate students' needs and future career development in the era of generative AI. In a diverse classroom with varying backgrounds, LLMs can be used effectively to accommodate all students' needs (Wen et al., 2024).

2. Methods

2.1. Preparing textbook homework question and answer key dataset

This study leveraged Bioprocess Engineering course (*EECE 506*) at Washington University in St. Louis. This core course is designed for graduate students from environmental engineering and chemical engineering. We used the text book "Bioprocess Engineering: Basic Concepts" (Third Edition) by Shuler and Kargi (2002) for our homework questions and corresponding answer key. Tested questions were selected from the end of each chapter and divided into smaller sub-questions for focused assessment. Each question was categorized into one of three types:

1. Conceptual Questions: These questions require explanations or brief descriptions about terminologies and concepts without numerical computations.
2. Numerical Calculations: These questions involve performing step-by-step mathematical calculations to arrive at a final numerical answer.
3. Equation Derivations: These questions require students to derive equations using theories taught in class.

2.2. Using LLM for solving textbook questions and grading

The textbook questions were solved using OpenAI API with two models: gpt-4o-2024–08–06 and o1-preview-2024–09–12. A standardized prompt: "Solve this problem as an exam problem. Be concise with your answer.", was used for all questions. For grading, two approaches were employed. First, a human grader evaluated the responses according to the answer key. Then second, both language models also graded the responses according to the answer key, and their scores were compared with the human graders to assess the consistency between human and AI evaluations. The grading scheme is either 0 or 1 point for each question, where 0 is incorrect final answer and 1 is correct final answer. To make questions more challenging for LLMs, we added redundant information, such as extra constants, equations, and data.

2.3. Real-world classroom exams with LLM

To assess the impact of LLMs on problem-solving, we designed in-class exams with two parts (both parts have the same questions), where 24 graduate students and 4 high school students participated in the exam. In Part A (40 minutes), students answer questions using only textbooks and class notes. In Part B (30 minutes), they can use GPT-4o (via the university's generative AI platform) to revisit questions they found challenging in Part A. Any response given in Part B overwrites the corresponding answer from Part A, regardless of correctness. After the exam, students completed a survey collecting data on their academic backgrounds. Besides human test takers, twelve LLMs were selected to solve the exam problems by authors. Nine of them are cloud-based: OpenAI o1, o3, GPT-4o, GPT-4.5, o3 mini, DeepSeek R1, DeepSeek V3, Qwen2.5 Max, and Qwen3–235B-A22B. Three are locally deployed on a PC with consumer-level GPU: DeepSeek R1 14B Qwen Distilled, Qwen QwQ, and Qwen2.5 14B. Then a human grader grades all responses according to an answer key.

2.4. Statistics test

The statistical analyses were conducted using Python. A series of propositions were tested to evaluate the impact of LLM assistance on exam performance. Pearson's correlation coefficient (r) was computed to assess linear relationships between key variables. These included LLM usage vs. overall performance, proportion of positive score gains vs. LLM usage, and Part A exam baseline score vs LLM score gain. One-sample t-tests were conducted to determine whether the mean score gain (or question-specific gain) significantly differed from zero. To further compare performance across subgroups, an independent-samples *t*-test was used. Exam participants were divided into two groups based on whether they have undergraduate biology background. The average score gains of the two groups were compared. This analysis aimed to reveal whether students with different baseline performance levels benefited differently from the LLM assistance. Additionally, Levene's test was applied to assess the variances in total scores between groups. Throughout the analyses, all tests were two-tailed. The underlying assumptions for each test, such as normality for t-tests and linearity for Pearson correlations, were considered during the analysis.

3. Results and discussion

3.1. LLM's ability of solving textbook questions

Since the emergence of LLMs, one of the concerns is that they will change education permanently, because LLMs can quickly solve homework/exam questions and give students an unfair advantage. In fact, solving numerical problems and writing essays have been key benchmarks for evaluating LLM capabilities from the beginning. When GPT-4 was launched in March 2023 (OpenAI, 2023), its groundbreaking feature was its ability to pass standardized tests like the SAT, bar exams, and MCAT by outperforming more than 90 % of human test-takers. In this study, we focus on graduate level bioprocess engineering education scenarios. We used OpenAI LLMs, o1 and GPT-4o, to solve textbook homework questions in each chapter of a bioprocess engineering textbook (Table 1). In a typical course setting, the most common types of homework questions are conceptual questions, numerical calculation questions, and equation derivations. The distribution of these question types in the textbook is shown in Fig. 2a, with numerical calculation questions being the most. In chapters covering biological basics, cell functions, metabolic pathways, and genetic alterations, most questions are conceptual. Both LLMs demonstrated exceptional proficiency in answering these conceptual questions, achieving near 100 % accuracy. This is not unexpected because scientific Q&A is a core benchmark for LLMs, and developers have significantly enhanced LLM's factual accuracy.

Table 1
LLM performance on bioprocess engineering exam.

LLM	Description	Exam 1 score	Exam 2 score
OpenAI o1	OpenAI Chain-of-thought reasoning LLM. It can read both text and image. It is closed-source and proprietary.	89 %	79 %
OpenAI o3	The latest OpenAI Chain-of-thought reasoning LLM. It can read both text and image. It is closed-source and proprietary.	90 %	84 %
DeepSeek R1	The most powerful DeepSeek Chain-of-thought reasoning LLM. It can only read text. It is open-source and free to download.	85 %	84 %
OpenAI o3 mini	OpenAI Chain-of-thought reasoning LLM. It can read both text and image. It is closed-source and proprietary.	78 %	72 %
Qwen2.5 Max	Alibaba AI lab's GPT LLM without Chain-of-thought reasoning. It can only read text. It is closed-source and proprietary.	78 %	71 %
Qwen QwQ 32B	Alibaba AI lab's Chain-of-thought reasoning LLM. It can only read text. It is open-source and free to download.	74 %	76 %
Qwen3–235B-A22B	The latest Alibaba AI lab's Chain-of-thought reasoning LLM. It can only read text. It is open-source and free to download.	87 %	86 %
DeepSeek V3	DeepSeek's GPT LLM without Chain-of-thought reasoning. It can only read text. It is open-source and free to download.	73 %	67 %
DeepSeek R1 14B Qwen Distilled	Qwen2.5–14B with post-training distillation by DeepSeek R1 data. It can do Chain-of-thought reasoning. It can only read text. It is open-source and free to download. This model can be easily deployed on personal computer.	70 %	48 %
OpenAI GPT−4o	OpenAI's GPT LLM without Chain-of-thought reasoning. It can read both text and image. It is closed-source and proprietary.	68 %	71 %
OpenAI GPT−4.5	The latest OpenAI's GPT LLM without Chain-of-thought reasoning. It can read both text and image. It can call external Python coding tools to solve mathematical problems. It is closed-source and proprietary.	75 %	85 %
Qwen2.5 14B	Alibaba AI lab's GPT LLM Qwen2.5–14B without Chain-of-thought reasoning. It can only read text. It is open-source and free to download. This model can be easily deployed on personal computer.	51 %	54 %

However, when it comes to numerical calculation problems, the two LLMs showed inferior performance. GPT-4o correctly solved only 28 % of questions, while o1-preview, with its enhanced reasoning capabilities, solved 43 %. The common errors are mainly calculation errors and wrong equation application. Calculation errors stemmed from flawed chain-of-thought, often resulting in answers being off by a factor of ten. Incorrect equations were due to the LLMs using equations outside the textbook context, leading to wrong final answers. Performance varied by chapter complexity; they performed better on simpler problems like cell growth calculations in Chapter 6 but struggled with complex enzyme kinetics equations in Chapter 3. When it comes to derivation of equations, the two models performed well. The probable reason is that the derived equations are present in LLM pre-training text. By analyzing LLM performance in real-classroom textbook questions, we reached the conclusion that the current state-of-the-art LLMs are not capable of helping students solve complex calculation problems. However, instructors should be cautious to assign conceptual questions and equation derivations.

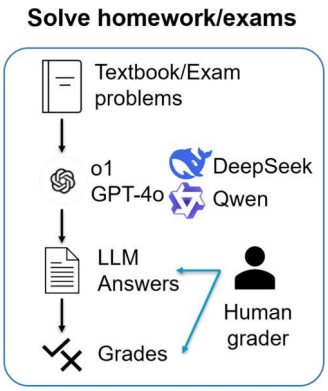


Fig. 1. Workflow for assessing Large Language Models (LLMs) in a graduate-level bioprocess engineering course. We evaluated OpenAI o1, GPT-4o, DeepSeek, and Qwen models in solving homework/exam questions.

3.2. *Insights from two real classroom exams*

To examine the impact of generative AI on student performance in a controlled classroom setting, we conducted an exam permitting AI/LLM use for the first time in our university. The exam comprised 15 multiple-choice questions (question 1–15), 8 short-answer questions (question 16–23), and 1 equation derivation (question 24). Numerical calculation questions were excluded due to time constraints. The average score for Part A (books and notes only) was 48.9 out of 100 (standard deviation 14.5). Allowing GPT-4o in Part B significantly improved student performance, raising the average score to 66.3 (standard deviation 11.1). Interestingly, increased AI usage correlated with lower overall scores (Fig. 3a).

A detailed analysis revealed that multiple-choice questions saw little improvement with AI assistance (Fig. 3**b**). In contrast, short-answer questions showed substantial score increases ($p < 0.001$). The highest gain was observed in the equation derivation problem on non-competitive enzyme inhibition. Additionally, in Fig. 3**c**, students with lower baseline score in Part A seem to benefit more from AI assistance ($r = -0.39$, $p = 0.061$). When analyzing the proportion of questions attempted with AI: a higher proportion of AI-assisted attempts corresponded to a lower success rate in score improvement ($r = -0.50$, $p = 0.012$). Although students gained more total score with more Part B attempts, those students who heavily relied on AI assistance still had lower total scores than those who did not. This data showed an urgent need for AI literacy which involves understanding not only how to use AI/LLM tools but also when and why to use them. Students need to learn to discern which tasks benefit more from AI assistance and critically evaluate LLM outputs.

After the first exam permitting LLM usage, students gained experience in formulating effective prompts and using LLM tools. We administered a second exam one month later. Students' usage strategy improved. In the first exam, students who used the LLM more tended to score lower, suggesting that they initially struggled with effective usage. However, in the second exam, their strategy improved (Fig. 4**a**), where there is a moderate positive correlation between LLM attempts and total score ($r = 0.40$, $p = 0.056$). Fig. 4**b** showed a similar positive trend, demonstrating that students were able to achieve, on average, positive gains across all questions. Fig. 4**d** highlights a strong correlation between total score gain and LLM usage attempts ($r = 0.78$, $p = 0.000$), a notable improvement compared to the first exam. LLM usage requires practice, and the more students engage with it, the better they become. To prepare students for future careers, it is essential to integrate AI literacy and practical application into higher education.

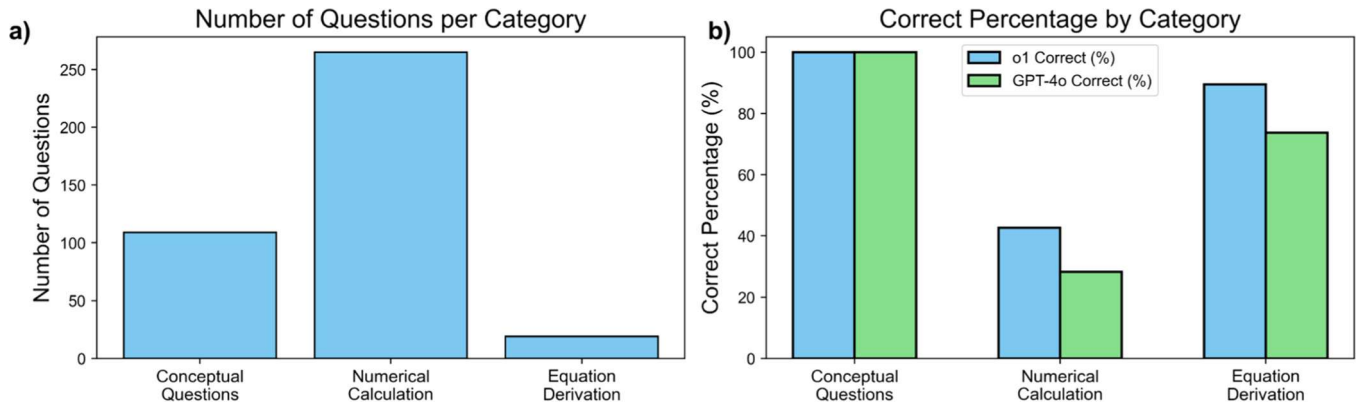


Fig. 2. Question category and LLM's accuracy. (a) Number of questions per category. Category 1 is conceptual understanding questions. Category 2 is numerical calculation questions. Category 3 is equation derivation. (b) Correct percentage by category for two models (o1 and GPT-4o).

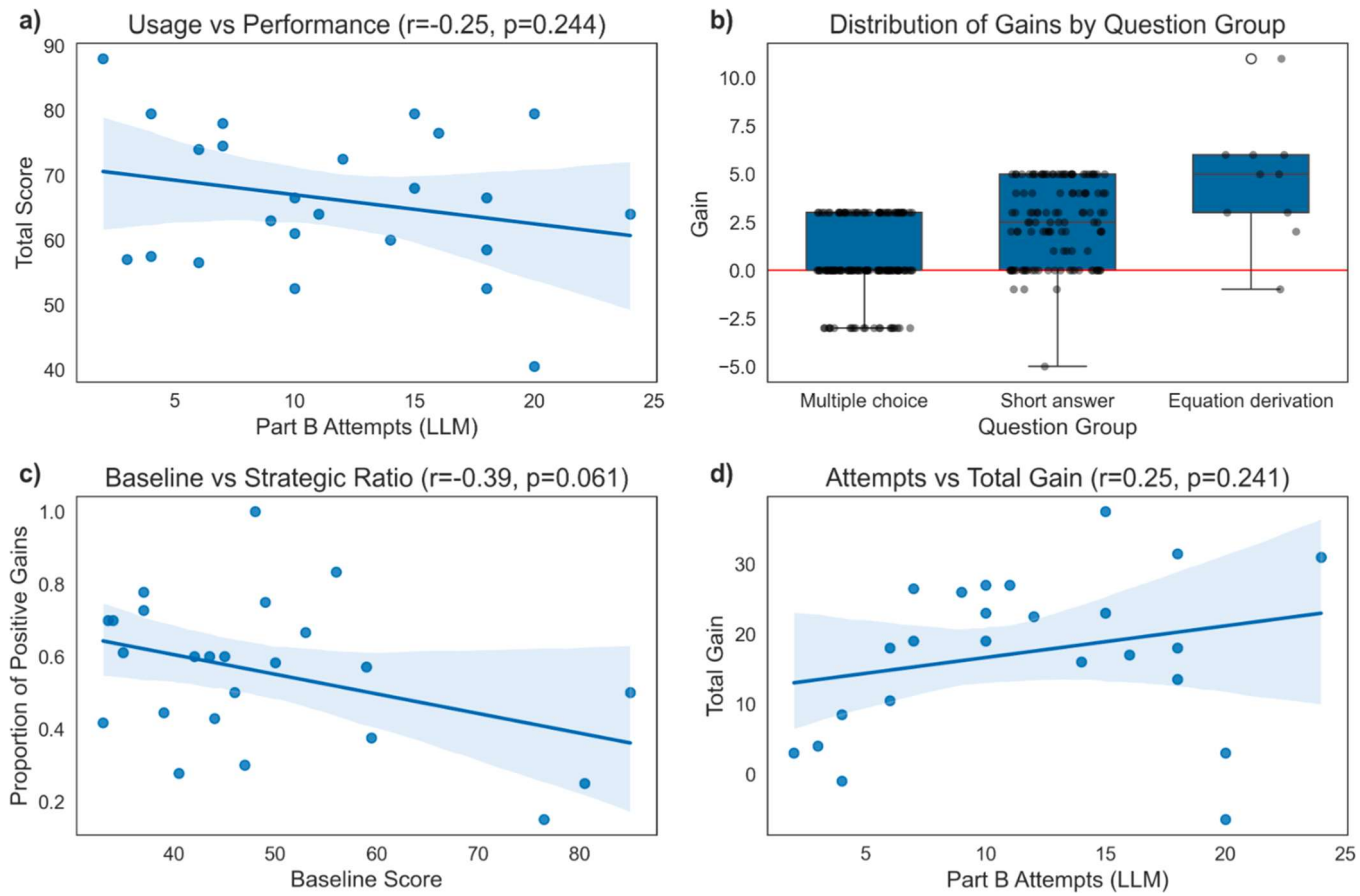


Fig. 3. Analysis of student scores in Exam 1. (a) Relationship between Part B attempts using LLM and total score out of 100. (b) Average score gains in each question group, with a red line indicating zero gain. (c) Proportions of questions with positive score gain in Part B (strategic ratio) versus baseline score in Part A. (d) Correlation between Part B attempts using LLM and total score gain. Note: shaded envelopes represent 95 % confidence intervals in linear regressions.

3.3. Impact of LLM usage on students with different backgrounds

Post-exam survey indicated that 15 students had an undergraduate background in biology, while 9 did not. To investigate the impact of prior biology knowledge on exam performance and AI usage, we analyzed key performance metrics across these groups in Exam 1. Students without a biology background had a significantly lower baseline score in Part A (39.78 vs. 54.33, $p = 0.014$). However, they exhibited a greater average score gain per question with AI assistance (2.33 vs. 1.33, $p = 0.016$) and a higher total score gain (25.56 vs. 12.47, $p = 0.003$). Those students without a biology background also had a higher successful score gain ratio in AI-assisted attempts (0.69 vs. 0.48, $p = 0.011$). Despite these differences in score improvement, there was no significant difference in total exam scores between the two groups (65.33 vs. 66.80, $p = 0.763$). Similarly, AI usage patterns, including the number of AI-assisted attempts (12.00 vs. 11.40, $p = 0.823$) and overall usage ratio (0.50 vs. 0.48, $p = 0.823$), did not differ significantly. Score variance was also comparable (4.34 vs. 3.95, $p = 0.708$). These findings suggest that while students without a biology background benefited more from AI in terms of score improvement, prior subject knowledge

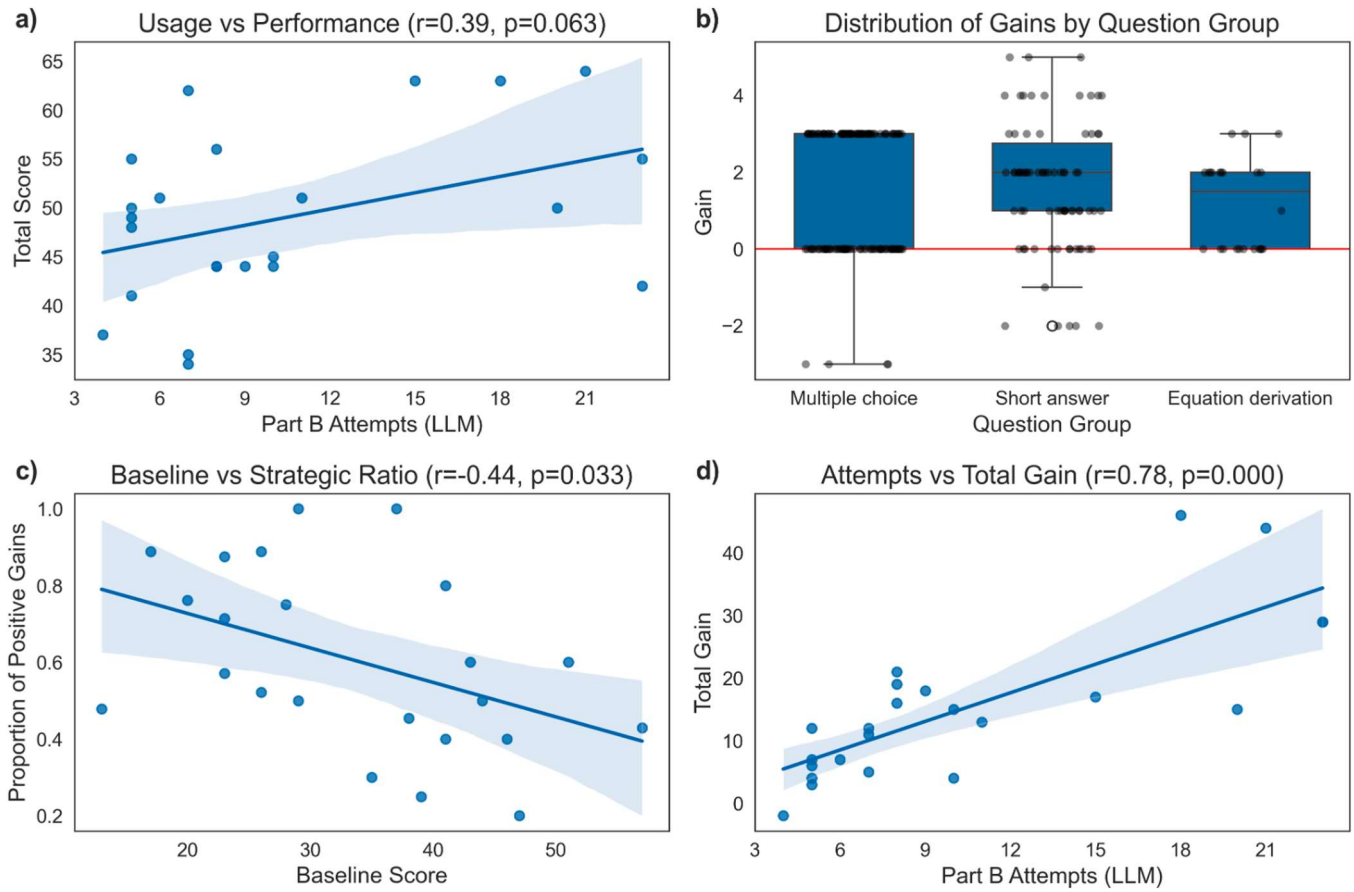


Fig. 4. Analysis of student scores in Exam 2. (a) Relationship between Part B attempts using LLM and total score out of 85. (b) Average score gains in each question group, with a red line indicating zero gain. (c) Proportions of questions with positive score gain in Part B (strategic ratio) versus baseline score in Part A. (d) Correlation between Part B attempts using LLM and total score gain. Note: shaded envelopes represent 95 % confidence intervals in linear regressions.

remained a key factor in baseline performance in Part A. The similar final scores indicate that AI assistance may help bridge the gap between students. Two students with cognitive disabilities with 50 % extra time accommodation achieved above-average exam scores with the help of LLM. High school students can achieve an average score of 69 ± 3 ($n = 4$) with sufficient LLM assistance, outperforming most graduate students. However, even graduate students with a background in biology may struggle to identify errors in LLM-generated responses without proper education in LLM literacy. As AI tools become increasingly capable of delivering near-perfect answers, there is a growing need to reconsider academic qualifications.

3.4. *LLMs in zero-shot exam settings: performance, equality, and security*

In addition to student test takers, nine large language models (LLMs) were engaged in the exam under best practices by an experienced user. Nine of these were cloud-based models—OpenAI o1, GPT-4o, GPT-4.5, o3, o3 mini, DeepSeek R1, DeepSeek V3, Qwen2.5 Max, and Qwen3.5–235B-A22B—while three, DeepSeek R1 14B Qwen Distilled, Qwen QWQ and Qwen 2.5 14B, were deployed locally on a consumer-grade GPU (Table 1). The LLM outputs were directly graded without any processing. However, this study does not aim to provide a rigorous LLM benchmark. LLM outputs are inherently stochastic, and various parameters—such as quantization, temperature, top-k sampling, and context length—can significantly influence performance. Our result is a typical use case under zero-shot condition. This disparity among LLMs can lead to social inequality where economically privileged students can gain more from paid version of AI/LLM tools. Notably, the locally deployed DeepSeek R1 14B and Qwen QwQ achieved scores comparable to some cloud-based models, likely due to recent advances in model distillation and chain-of-thought reasoning. Even a relatively small model like Qwen QwQ can now surpass the class average, suggesting that offline exams without internet access may be compromised by students using local LLM on portable laptops with GPU (or Mac with unified memory). This may lead to social inequality, where students with financial advantage can pay for better computation hardware to win.

3.5. *Designing difficult questions for LLMs*

Through constructing real-classroom homework dataset and systematic analysis, we got some ideas about LLM's behavior when solving textbook questions. Now one question arises, how can instructor design questions that are difficult for LLMs. We analyzed their performance on real-classroom homework datasets and found that LLMs sometimes struggle with redundant information in question statements. By adding irrelevant details such as extra constants and equations in the question statements, o1's correct rate was reduced by 31 %. Also, given that LLMs lack specific course context, instructors can incorporate recent research and publications. This approach encourages students to engage deeply with the literature and course material rather than paying attention to LLM-generated contents. In the class exam, LLMs exhibited several common errors. We summarized our findings in Table 2.

LLMs occasionally confuse similar terms, such as ribulose and ribose (as the key component of DNA), indicating that subtle wording changes can exploit weaknesses in vector-based word embeddings. LLMs frequently isolate single factors while neglecting interdependent variables. In a short answer question, students were expected to recognize

Table 2
Performance of LLMs in different types of questions.

Question type	Performance	Example bioprocess engineering question
General conceptual understanding	Good	What are the main pathways in cell's central metabolism? LLM Answer: Glycolysis, pentose-phosphate pathway, and TCA cycle.
Information about one entity	Fair	Describe potential problems to use formamide as nitrogen source for growing *E. coli*. LLM incorrectly states that formamide is toxic to *E. coli*, despite active research is going on with this novel nitrogen source.
Question requires critical thinking	Poor	For a lactate dehydrogenase catalyzed reaction, the enzyme reaction follows Michaelis-Menten equation. Michaelis constant is 0.1 mM and lactate concentration is 2 g/L. The initial cofactor NAD is 15 mmol. The maximal enzymatic reaction rate coefficient Vmax is 0.2 mM/second. How much time is required to convert 90 % of lactate to pyruvate? Most LLM ignore the fact that there is not enough cofactor NAD to accomplish this reaction.
Graphical pathway interpretation	Poor	LLM incorrectly identified cAMP as AMP.
Negatively worded conceptual biological questions	Poor	Which of the following is not true about the enzyme kinetics? *A. The higher the kcat is, the higher the turnover rate of the enzyme is.* *B. The enzyme rate cannot be faster than the diffusion-limited encounter of an enzyme and its substrate.* *C. A larger Km is an indicator of high affinity for enzyme substrate binding.* *D. When substrate concentration is much smaller than Km, the rate of enzyme reaction is highly impacted by the concentration of the substrate.*

that the culture medium lacked both a nitrogen and carbon source, making it unsuitable for sustaining *E. coli*. However, LLMs focused only on the carbon source, overlooking the nitrogen limitation. LLMs often make error about some highly specific domain knowledge in bioprocess engineering literature. For example, they misidentified formamide (a novel nitrogen source) at 1 g/L as highly toxic to *E. coli*. LLMs also failed to reconcile theoretical biochemical knowledge with real-world feasibility. In a short answer question, LLMs incorrectly suggested base saponification for biodiesel production instead of the acid- or enzyme-mediated hydrolysis approach. This generation of Chain-of-thought LLMs are capable of deriving equations and solving calculation problems, but they still struggle with complex calculation with graphs and data tables. In one question asking students to calculate reaction time required, nearly all LLMs failed to identify the insufficient NAD present. It seems that LLMs are not capable of reasoning through the paradox of the question. Overall, the main limiting factor preventing students from taking advantage of LLMs usage is exam time limitation, because Chain-of-thought LLMs require longer reasoning time.

3.6. Feasibility of LLM as grader

One of the time-consuming works in higher education is grading student homework, and it is more difficult to hire teaching assistants these days due to lack of qualified candidates. We explored the feasibility of LLM as grader. When given answer keys, GPT-4o grader and o1 grader had average discrepancies of 46 $\pm$ 1 % and 27 $\pm$ 1 % compared to a human grader. LLMs tend to assign scores rather than clearly label answers as incorrect. Also, students may make small markings to assist their thinking process, and when LLM reading scanned exams, LLM's attention mechanism cannot attend to all the markings. At this stage, LLMs are not reliable graders, and human supervision is strongly recommended if they are to be used in this role.

3.7. Class survey about AI/LLM usage

94 % of the students on the course found ChatGPT to be more accessible and helpful throughout the course to assist with homework and exams (Fig. 5). Other AI/LLMs used were Copilot and Deepseek. While most students didn't require AI to solve excel calculation problems, they used AI to solve computer programming problems. Many students reported that writing an effective prompt to get a precise answer requires significant time and effort. It also demands fundamental knowledge of the subject to ask the appropriate questions and to proof-check the answer. Learning the skills to format the right prompt requires practice. Encouraging the use of AI can help students practice such skills. Therefore, learning to use AI effectively can be useful for further coursework and future career. It was further mentioned that not all AI models can process information with equal efficiency, and that it would be fairer if everyone could use the same model. This highlights the potential inequality faced by students who cannot pay for advanced AI models, which may put them at significant academic disadvantage over students who can afford expensive models. To promote equal opportunities in academic performance, institutions should provide all students with access to advanced AI models, as this would eliminate the disparities in performance caused by limited access to the advanced AI models.

3.8. MATLAB coding tasks in bioprocess engineering course

In bioprocess engineering course, one topic is on simulation of time course bioreactor fermentation data by solving ordinary differential equations. Chemical engineering education curricula has emphasized basic coding skills in MATLAB/Simulink and Python. MATLAB prepares students for process simulation and differential equation solving, while Python is essential for data-driven studies and machine learning. However, with the advent of coding LLMs like Microsoft Copilot, ChatGPT and Qwen code, basic coding courses have become less relevant. Students can now write code using natural language instructions. For example, our course had a MATLAB assignment for simulating two-species predator-prey population dynamics, and students took advantage of GPT-4o for generating correct code instantly. Such assignments, typically given as take-home homework, now need to be reconsidered. Instructors should instead demonstrate LLM code generation and encourage students to pursue higher-level coding objectives.

3.9. Concluding remark: promoting AI literacy

This study reveals both the transformative potential and limitations of LLMs in bioprocess engineering education. Classroom exams demonstrated that LLM access improves student grades over time, but LLMs still struggle with long numerical calculations. Equality concerns emerge from disparities in LLM quality, as premium models outperform free alternatives, thereby granting academic advantages for financially privileged students. AI/LLM's vulnerability to contextually rich, critical-thinking questions suggests that curricula must evolve to emphasize higher-order skills and integrate AI literacy. To ensure that students develop practical skills, we designed a lecture module introducing LLMs within the bioprocess engineering course (See GitHub page https://github.com/xiao-zhengyang/EECE506-SP2025). These lectures cover the history of LLMs, key terminologies, and fundamental concepts necessary for understanding LLM capabilities and limitations. In addition, we incorporate hands-on activities where students learn how to interact with LLMs across different platforms (online or on a personal computer). A key component of this module is a live demonstration of GraphRAG, an advanced retrieval-augmented generation (RAG) system that reads

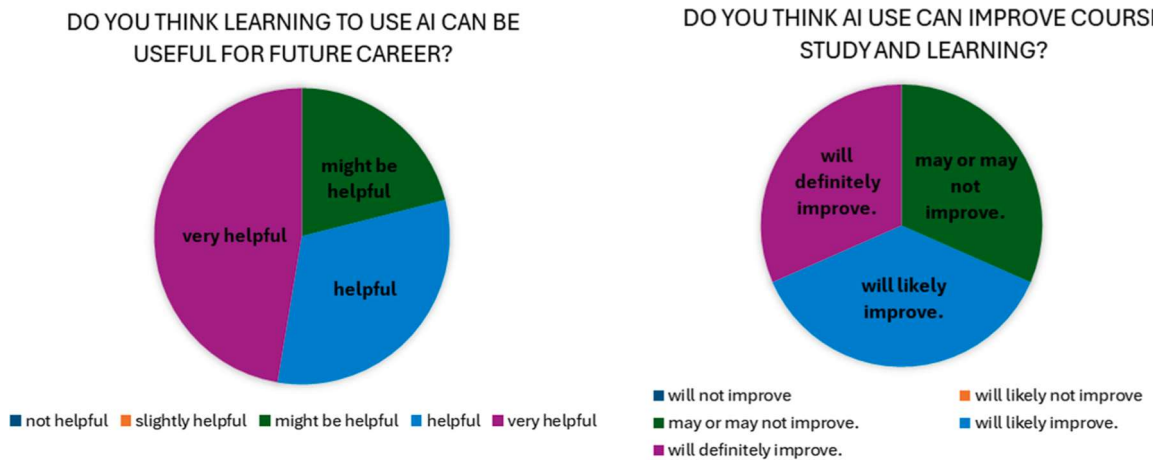


Fig. 5. Class survey of AI/LLM usage.

text and organizes into knowledge graphs for better information retrieval (Edge et al., 2024). These activities equip students with essential computational skills, bridging AI learning and real engineering applications, and preparing them for the future job in synthetic biology and bioprocess engineering.

CRediT authorship contribution statement

Zhengyang Xiao: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis. **Eunseo Lee:** Writing – review & editing, Writing – original draft, Data curation. **Tang Yinjie:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Sophia Yuan:** Data curation. **Roland Ding:** Data curation.

Disclaimer

The analysis and conclusions presented herein are all based on publicly available information and aim to contribute to ongoing academic conversations about responsible AI/LLM usage and its implications for higher education. The views expressed in this article are solely those of the author and do not represent the official position or opinions of any organization, institution, or employer associated with the author. This article is intended for academic discussion purposes only and should not be interpreted as an endorsement or critique of any specific policies, entities, or jurisdictions.

Declaration of Competing Interest

The authors declare no conflict of interest.

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